# Comparative Evaluation of Simple CNN, POSTER_V2, and LibreFace for Facial Expression Recognition on the RAF-DB Dataset

## I. Introduction:

Within the broader domain of computer vision, image classification serves as a fundamental benchmark for evaluating how effectively deep neural networks parse semantic visual content [1]. According to [2], Facial Expression Recognition (FER) operates as a highly specialized sub-discipline focused on non-rigid, subtle, and highly dynamic structural variations, which map dynamic changes in facial muscles to psychological states. Therefore, FER acts as a critical technical bridge between low-level pixel data and high-level human affective computing [3].

Despite major structural advancements in deep convolutional networks, robust automated FER remains highly challenging due to several distinct environmental and physiological constraints:

i. **Intra-class Variability:** Expressive variance within the same emotional category is exceptionally high [4]. This is because factors such as demographic diversity, micro-expressions, cultural differences in emotional manifestation, head pose variations, and non-uniform illumination fields mean that the same underlying emotion can be present differently.
ii. **Inter-class Similarity:** Different emotional states often share highly overlapping facial configurations, creating highly complex, non-separable feature boundaries in Euclidean latent space [5] [6].
iii. **Data Imbalance:** Real-world data is naturally skewed, where multiple emotions are blended [7]. Furthermore, natural data distribution skews bias deep optimization pipelines heavily toward majority emotion classes [8].

To address these challenges, this project conducts a rigorous comparative study across three distinct architectural paradigms. Models are optimized, validated, and evaluated on the real-world affective faces database. The baseline architectures selected for this benchmark span different levels of structural complexity:

i. **Simple CNN**: A lightweight, custom-built Convolutional Neural Network serving as a structural baseline to isolate raw feature learning without advanced abstractions.
ii. **POSTER V2**: A state-of-the-art FER architecture that leverages multi-scale feature abstraction and modern transformer-based attention mechanisms to isolate fine-grained facial variations.
iii. **LibreFace**: A modern, comprehensive FER toolkit designed for holistic facial feature extraction and optimized cross-domain representation.

Thanks to enforcing a standardized prediction interface, this framework guarantees identical metric computation across all configurations.

The report is organized as follows:

i. Section 2 details data preparation, preprocessing steps, and the standardization of the 7-class emotion mapping.
ii. Section 3 delineates the architectural paradigms and structural configurations of the three evaluated networks.
iii. Section 4 outlines the experimental setup, hyperparameter configurations, and training environments, and delivers a thorough analysis of empirical results.
iv. Section 5 summarizes our core findings and presents perspectives on future optimization paths.

## II. Background Theory:

Two popular approaches used for analyzing facial expression are attention-based and facial Action Unit (AU) intensity estimation [9]. The attention-based method focuses on capturing the face location to indicate the predicted region. In contrast, AU intensity is introduced as a process to qualify the strength of facial movement based on the degree of deformation and muscle placement. The intensity usually is analyzed under the scale 0

to 5 in various aspects [9]. Therefore, instead of predicting the whole arbitrary image, attention-based methods improve the feature selection and regional focus, while the AU-based approach provides descriptive information of facial expression.

As FER deals with image data, Convolutional Neural Network models are preferred because of their spatial learning ability. The idea was introduced by LeCun et al. (1998) [10], who showed that stacked convolutional layers can learn patterns directly from pixels. A convolutional layer slides small filters across the image to detect local features such as edges, and because each filter is reused everywhere, it needs far fewer parameters than a fully connected layer. When several layers are stacked, the early ones learn simple features and the deeper ones combine them into more complex facial patterns.

Each block in our model also uses batch normalization, *ReLU* activation, and max pooling. According to [11], batch normalization normalizes the inputs of each layer to make training faster and more stable. *ReLU* adds nonlinearity, and max pooling shrinks the feature maps to lower the computation and add some tolerance to small shifts.

Instead of flattening the final feature map, our model uses global average pooling [12], which gives a compact feature vector no extra parameters and less overfitting. Dropout [13], which randomly switches off units during training, so the model does not rely too much on any single feature. Finally, the RAF-DB dataset [14] is naturally imbalanced across seven emotions, which helps explain why the model learns some emotions better than others.

However, CNNs may struggle in real-life situations where facial images could be affected by pose, lighting, exposure, etc. To overcome these problems, a combination between encoders and CNNs is suggested to maintain the robustness and accuracy of the system. This suggested solution is shown in the deployment of POSTER_v2 with landmark-guided attention based and LibreFace with AU-based.

In the following LibreFace section, there exists a ResNet-18 model. The ResNet-18 [15], short for Residual Neural Network with 18 layers, is a strong deep-learning architecture to solve the degradation by adding residual blocks with shortcut connections. In LibreFace, it is used as a lightweight student model through the feature-wise knowledge distillation from teacher model Masked Auto-encoder (MAE) [16] to provide a practical balance between performance and efficiency for real-time FER.

# III. Dataset and Evaluation Methodology:

The Real-world Affective Faces Database, RAF-DB [14], is a facial expression dataset that takes images from the internet. It was created for facial expression recognition research in unpredicted conditions, between age, gender, ethnicity, lighting, head pose, occlusion.

The original RAF-DB database contains 29,672 images with 100×100 pixels and includes both a single label subset and a compound emotion subset. In this project, we used the single label subset, which contains 15,339 images divided into seven classes. This subset split contains 12,271 training images and 3,068 test images. The original training set was divided into 9,816 training images and 2,455 validation images with the ratio 4:1.

| Class_id | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Class_name | Surprise | Fear | Disgust | Happiness | Sadness | Anger | Neutral |
| Count | 1619 | 355 | 877 | 5957 | 2460 | 867 | 3204 |
| Percentage | 10.55% | 2.31% | 5.72% | 38.84% | 16.04% | 5.64% | 20.89% |

*Table 1 - Class Distribution of RAF-DB dataset*

It is necessary to use a unified evaluation framework to compare three models equally. With the CSV format: image*_path*, true*_label*, pred*_label*, *prob_1*, *prob_2*, *prob_3*, *prob_4*, *prob_5*, *prob_6*, *prob_7*, and class mapping is the same for all models. The results based on prediction of CSV were used to calculate the evaluation metrics, which are accuracy, F1 score per class, Weighted, Macro F1, and confusion matrix.

Since the class is imbalanced, accuracy is not the only metric we need. Fear, Disgust, and Anger have a very low number of images compared to others. Therefore, we need macro F1 and weighted F1. Finally, a confusion matrix checks the correct and incorrect prediction for each class, so we can also see the impact of class imbalance on the classification.

The advantages of RAF-DB are that it contains real Internet images with diversity and random images patterns. So, it is more realistic than cleaning and setting up datasets. The last one is that the dataset was annotated through crowdsourcing. Therefore, it can improve label reliability compared to a single annotator. However, RAF-DB also has limitations; resizing the images can also remove some image details. Some emotion classes are similar. For example, Fear and Surprise, also Disgust and Anger.

# IV. Models, Solutions, Implementation, and Result:

## 1. Proposed Solution:

To solve FER challenges namely intra-class variability, visual proximity and class imbalance, there are three architectural paradigms. Simple CNN was built from scratch with no pre-training and no attention. Visual overlap needs spatial precise distinguish from POSTER_V2's pyramid face-feature-guided cross-attention at three scales [17]. For LibreFace, the freeze-then-unfreeze fine tuning to RAF-DB label space is used based on AffectNet-pretrained teacher [18].

All three are evaluated under the same measurement conditions with shared CSV schema, train/test split, 224x224 ImageNet-normalized preprocessing. The accuracy, per-class F1, macro F1 metrics, and confusion matrices are used to evaluate the model's performance.

## 2. Simple CNN:

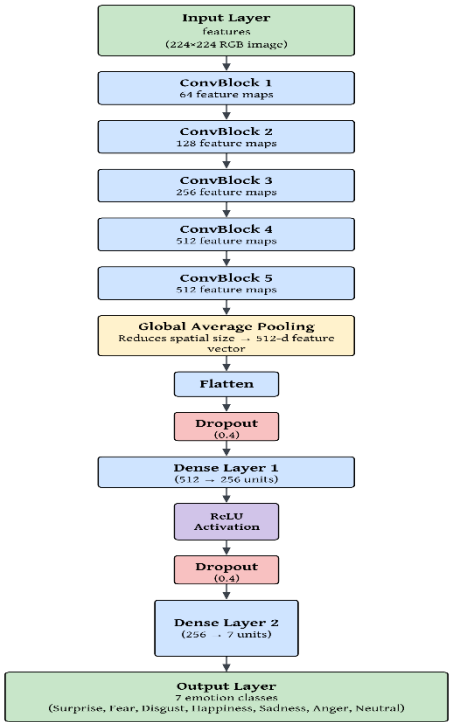



*Figure 1 - High Level Architecture of Simple CNN Model*

The Simple CNN, shown aside, takes an RGB image resized to 224x224 and has two parts. The feature extractor is a stack of five convolutional blocks, each applying a 3x3 convolution, batch normalization, a *ReLU* activation, and a 2x2 max pooling layer. The channels grow across the blocks (3, 64, 128, 256, 512, 512) while the spatial size halves at each step (224 down to 7). The final 512x7x7 feature map is reduced by global average pooling to a 512-dimensional vector, which keeps the model small. The classifier head then applies dropout, a linear layer (512 to 256), a *ReLU*, dropout, and a final linear layer (256 to 7), giving 7 logits.

The pipeline runs in three stages, training, inference, and evaluation. The model is trained with the *AdamW* optimizer and a cross-entropy loss with label smoothing, using data augmentation, a learning rate scheduler, and early stopping, and the best checkpoint is saved. In inference, this checkpoint produces *softmax* probabilities on the test set, and the highest probability class is the prediction. The output is written to "simple_cnn.csv", which is checked by the consolidation script and passed to the results generator for the metrics and confusion matrix.

On the RAF-DB test set of 3068 images, the model reached 76.14% accuracy, a macro F1 of 65.86%, and a weighted F1 of 75.80%. Happiness scored highest at 88.43% F1, while Fear and Disgust scored lowest at 46.72% and 45.48%. This pattern mostly reflects the class imbalance in RAF-DB, where Happiness has many training images, and Fear and Disgust have few. A lower score here does not mean the design is flawed; it mainly reflects how much data was available for each emotion. The gap between the macro F1 and the weighted F1 supports this, since the macro score weighs every class equally and is pulled down by the weak minority classes.

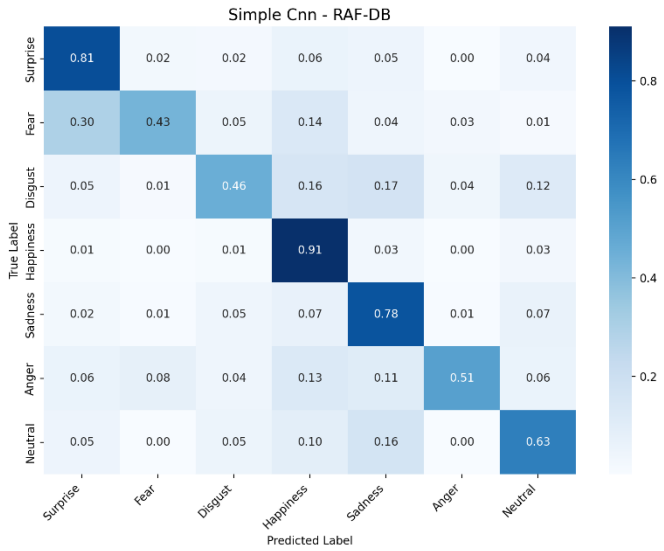



*Figure 2 - Confusion Matrix of Simple CNN Model*

The confusion matrix shows that the model performs best on Happiness (0.91) and Surprise (0.81), where the diagonal values are strong, and weakest on Fear (0.43), Disgust (0.46), and Anger (0.51). The largest value excluding diagonal values is 0.30, where Fear is predicted as Surprise, which makes sense because the two expressions look visually similar. Overall, the diagonal stays clearly dominant for every class, so the model is learning each emotion, just with lower confidence on the minority classes that have less training data.

## 3. POSTER_V2:

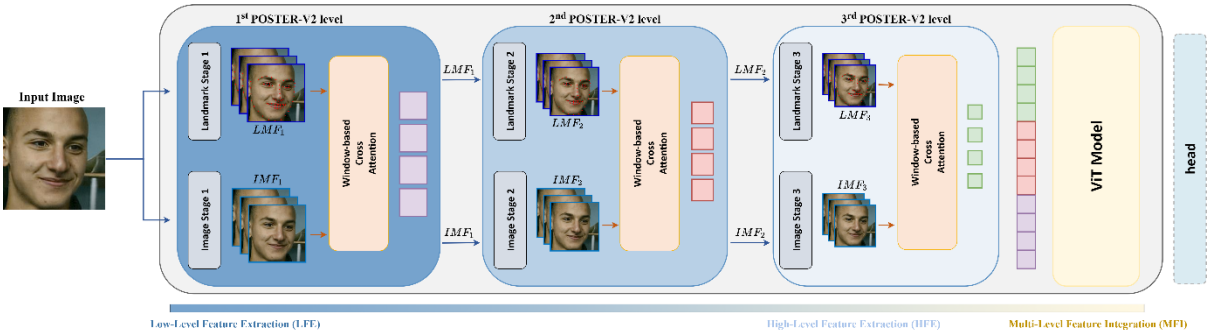



*Figure 3 - High Level Architecture of POSTER_V2 Model*

POSTER_V2 [17] is a dual-stream pyramid transformer. It fuses face recognition features (one stream) with image appearance features (another stream) through three-level architects. Images at 224x224 are processed by IR-50 to get feature maps at three levels: 28x28, 14x14 and 7x7. In parallel, the same image input is resized to 112x112 to pass through *MobileFaceNet* Encoder. Face-specific queries are generated after training with fixed weights (*requires_grad=False*). This is the key design decision to differ from this stream from a second trainable encoder. At each pyramid level, a *WindowAttentionGlobal* module uses *MobileFaceNet* and IR-50 outputs as keys. Head counts scale with depth: 2 at 28x28, 4 at 14x14, 8 at 7x7. All outputs are

projected to 768 dimensions and applied to a lightweight *ViT* (*depth = 2*, *embed_lim = 768*) that produces the final 7-class logits.

The model was adapted from the official repository for RAF-DB evaluation. Three pretrained weight files are required: the RAF-DB POSTER_V2 checkpoint, IR-50 face recognition weights, and *MobileFaceNet* weights. inference_wrapper.py remaps 0-indexed class labels to the project's 1-indexed convention, keeping predictions consistent with the shared CSV schema. Training used SAM ($\rho = 0.05$) wrapping Adam at $lr = 3.5 \times 10^{-5}$, with *ExponentialLR* ($\gamma = 0.98$) over 200 epochs and batch size 144. SAM executes two forward-backward passes per update: the first finds the sharpness-maximizing weight perturbation; the second updates parameters toward a flatter loss region (Foret et al., 2021). Augmentation applied random horizontal flipping and random erasing ($p = 0.5$).

The official repository from GitHub is cloned and adapted for this experiment. Training used SAM ($\rho = 0.05$), Adam at $lr = 3.5 \times 10^{-5}$, *ExponentialLR* ($\gamma = 0.98$) over 200 epochs and batch size 144. SAM requires two forward-backward passes per update. The first step finds the sharpness-maximizing perturbation, the second one finds flat-minima point [23]. Augmentation applied random horizontal flipping and random erasing ($p = 0.5$).

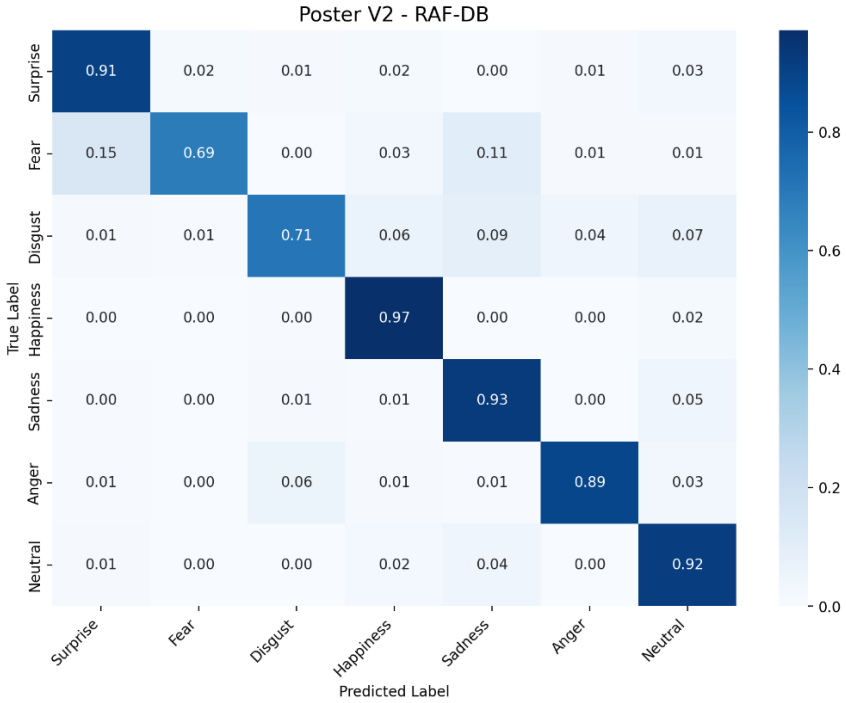
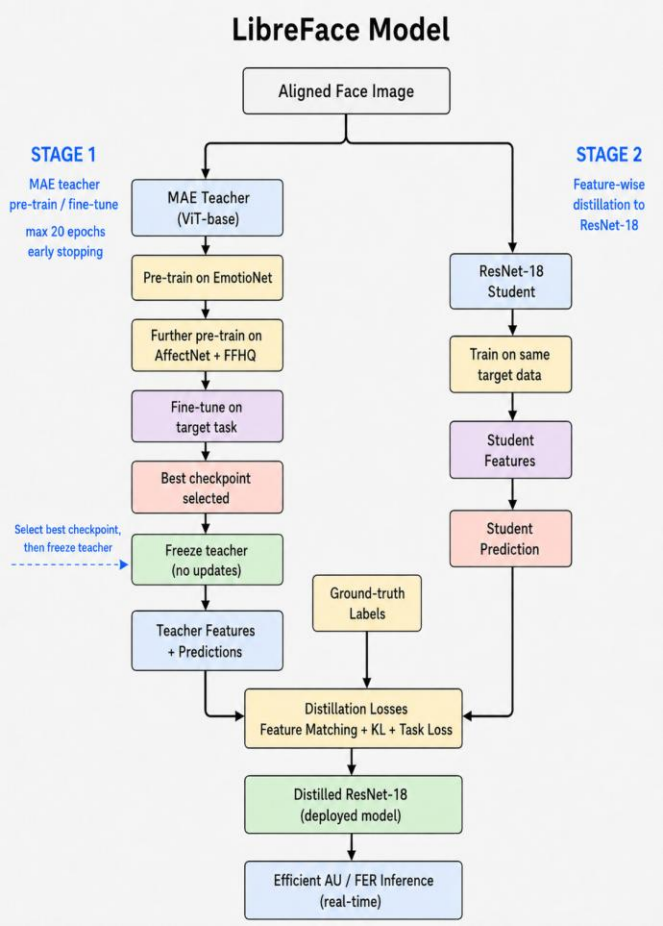



*Figure 4 - Confusion Matrix of POSTER_V2 Model*

The approach achieved 92.18% accuracy, 87.34% macro F1, and 92.09% weighted F1. Per-class F1: Happiness 97.05%, Neutral 91.19%, Surprise 90.85%, Sadness 90.70%, Anger 89.72%, Fear 76.12%, Disgust 75.75%. Due to having most training samples and different facial expressions from all others, happiness gained the highest scores. The small gap between weighted F1 and accuracy shows that the design of model improved minority-class scores not just focus on majority classes.

## 4. LibreFace:

LibreFace pipeline consists of two steps: pre-processing and training. Prior to inputting the data into a training model, face mesh and landmark detection within an input image are implemented by using an open-source tool - mediaPipe[18]. By having the facial mesh, we could identify precisely the locations of facial properties, such as mouth, eyes, eyes-brown and nose. Then, use the facial landmark to crop and extract the facial-focused image before passing it to a prediction model.




The training is structured in two phases using either supervised or self-supervised learning approach. The architecture follows 'teacher-student' framework, where teacher is responsible for the first phase and guides the student in the second.

According to figure aside, teacher model is the ViT-base structural Masked Auto-encoder (MAE) which is pre-trained on EmotioNet [20][21]. The given weights of the ViT encoder from MAE are used as the backbone with the addition of a linear classifier for future training on dataset. For the first stage, by using large model (MAE), the researcher from [18] pose a strict rule for training 'teacher' model for a maximum of 20 epochs with early stopping and saving the best checkpoint. The stage 2 then freeze the 'teacher' and train the light-weight ResNet-18 - 'student' model through feature-wise distillation based on 'teacher' checkpoint. During this process, only 'student' feature is fine-tunned base on the frozen weights from 'teacher' model. After that, the KL [22] divergence loss is employed to minimize the discrepancy between teacher - student classifiers.

The models are trained with the *AdamW* optimizer, using the loss function MSE for AU intensity estimation and cross-entropy for Facial Expression Recognition.

In our project, we applied a similar strategy by cloning the model on GitHub, then re-training in local environment and saving the best checkpoint using RAF-DB dataset, then evaluating the model performance based on final 7-class logits. The training was halted at epoch 26 due to no improvement.

The confusion matrix by LibreFace on RAF-DB with accuracy 85.72, close to the report in [18]:

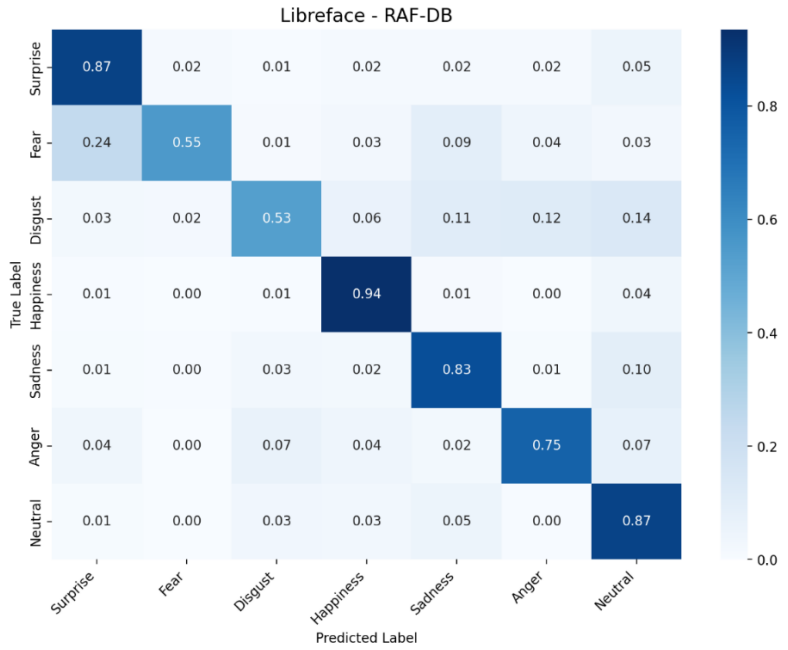



*Figure 5 - Confusion Matrix of LibreFace Model*

In general, LibreFace has a decent performance on 5 classes except “Fear” and "Disgust". This has resulted by the drawback of AU-based decomposition model as “Fear” expression shares the same facial properties with "Surprise", e.g.: upper eyelids pull, white part of eyes exposed, jaw dropped, etc. The similar pattern is also witnessed between “Disgust” and the last three classed.

## 5. Models comparison:

| Model | Accuracy (%) | Weighted F1 (%) | Macro F1 (%) |
|---|---|---|---|
| Simple CNN | 76.14 | 75.80 | 65.86 |
| POSTER_V2 | 92.18 | 92.09 | 87.34 |
| LibreFace | 85.72 | 85.62 | 77.58 |

*Table 2 - Summary metrics - RAF-DB test set (n = 3,068)*

| Model | Surprise | Fear | Disgust | Happiness | Sadness | Anger | Neutral |
|---|---|---|---|---|---|---|---|
| Simple CNN | 76.59 | 46.72 | 45.48 | 88.43 | 70.23 | 62.88 | 70.71 |
| POSTER_V2 | 90.85 | 76.12 | 75.75 | 97.05 | 90.70 | 89.72 | 91.19 |
| LibreFace | 86.97 | 64.06 | 55.74 | 94.30 | 82.93 | 75.54 | 83.53 |

*Table 3 - Per-class F1 (%) RAF-DB test set (n = 3,068)*

Fear and Disgust is the biggest gaps. Simple CNN scores closely on both (46.72% and 45.48%). Same pattern is witnessed in POSTER_V2 with 29.4 and 30.3 points respectively. LibreFace closes much of that gap but still trails 12.1 points behind POSTER_V2 on Fear and 20.0 on Disgust. Both feelings stimulate the muscles that pull down the brow. As shown in the confusion matrices, the cross-model misclassification is concentrated on these two categories.

Performance differences map to different deployment targets. The Simple CNN is suitable for running on CPU in real-time with accuracy is secondary concern. In opposite, when accuracy is priority, POSTER_V2 with 3 pretrained weight files brings experimenters desired result in GPU setting. For middle ground use if POSTER_V2’s dependencies are impractical, LibreFace should be taken into consideration with one standard ResNet-18 forward pass and one checkpoint only.

# V. Conclusion:

In conclusion, POSTER_V2 and LibreFace performed well on RAF-DB with high accuracy and had better results than the self-built Simple CNN model. Building a Simple CNN is useful as it helps us gain knowledge about the basic implementation of the Convolution Neural Network. For LibreFace, since the model does not have a checkpoint for RAF-DB, we trained the LibreFace model to retrieve the best checkpoint on RAF-DB. POSTER_V2 shows the best performance compared to the other 2 models, with the pretrain RAF-DB checkpoint. Through the model's comparison, the understanding of the importance of encoders is raised.

The confusion matrix shows that most of the classes are well-classified. However, Fear and Disgust is still challenging to predict for all the models, even Poster V2. This could be explained by the fewer images of them in the dataset and the similarity in facial patterns.

In future work, the given models would be deployed in real-time on a webcam to test their performance in real-word conditions. Since the blur, lighting changes, pose and other properties could lower the quality of image, a practical evaluation of model would be provided.

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